VBF-T2R1NOFORCE-CALC-001 - Design Calculation Sheet (F1)

Sheet: Inputs

Parameter	Value	Source
Base parameter set	Chen2020	entry 0 meta (anchor-table default; no electrode system in task text)
Design overrides (F1)	Negative particle radius 3e-6 m; Negative electrode porosity 0.42; SEI kinetic rate constant 4e-13	cell/params_f1.json
Positive electrode thickness / porosity / density	75.6 um / 0.335 / 3262 kg/m3	parameter set
Negative electrode thickness / porosity (F1) / density	85.2 um / 0.42 / 1657 kg/m3	parameter set + params_f1.json
Separator thickness / porosity / density	12 um / 0.47 / 397 kg/m3	parameter set
CC thickness Al / Cu	16 um / 12 um (2700 / 8960 kg/m3)	parameter set
Electrode height x width (area)	0.065 m x 1.58 m (0.1027 m2)	parameter set
Nominal cell capacity	5.0 Ah	parameter set Nominal cell capacity [A.h]
Voltage window	2.5 - 4.2 V	parameter set cut-offs
Cation transference number	0.2594	parameter set

Sheet: Capacity & Energy

Quantity	Formula	Value	Source
1C discharge capacity (25 C)	CC discharge at 5 A to 2.5 V	5.0282 Ah	cell/r5_f1_1c_spme.json:capacity_ah
-20 C discharge capacity	CC discharge at 5 A, 253.15 K ambient + initial temp (cold-start fix)	5.0003 Ah	cell/r5_f1_lowT_spme.json:capacity_ah
-20 C retention	100 x cap_lowT / cap_25C	99.45 %	derived/r5_f1_retention.json (mechanical)
Discharge energy (1C)	trapezoidal integral of voltage_v x I_1C dt / 3600	17.8485 Wh	cell/r5_f1_energy.json:energy_wh
Midpoint voltage	voltage at discharge-time midpoint	3.9794 V	cell/r5_f1_energy.json:midpoint_voltage_v

Sheet: Energy Density

Quantity	Formula	Value	Source
Layer masses	layer_kg_m2 x area (pos 0.16399, neg 0.08188, AlCC 0.0432, CuCC 0.10752, sep 0.002525) kg/m2 x 0.1027 m2	16.842 + 8.409 + 4.437 + 11.042 + 0.259 g	cell/r5_f1_energy.json:layer_kg_m2
Cell mass (electrolyte excluded)	sum of layer masses	40.990 g	cell/r5_f1_energy.json:mass_kg
Gravimetric energy density	energy_wh / mass_kg	435.44 Wh/kg (criterion >= 327.18: PASS)	cell/r5_f1_energy.json:energy_density_wh_kg
Volumetric energy density	energy_wh / (thickness x area)	865.50 Wh/L	cell/r5_f1_energy.json:energy_density_wh_l
DCR / power density	V_OC at t0 - V at 10% DOD over I_1C; V_OC^2/(4 DCR)/mass	0.2036 mOhm; 497.31 kW/kg	cell/r5_f1_energy.json:dcr_ohm,power_density_w_kg

Sheet: N-P & Mass

Quantity	Formula	Value	Source
Negative theoretical areal capacity	c_max 33133 mol/m3 x F/3600 x volfrac 0.75 x (1-0.42) x 85.2 um	32.91 Ah/m2	parameter set + params_f1.json

Quantity	Formula	Value	Source
Positive theoretical areal capacity	c_max 63104 mol/m3 x F/3600 x volfrac 0.665 x (1-0.335) x 75.6 um	56.54 Ah/m2	parameter set
N/P (theoretical, deliverable formula)	neg areal cap / pos areal cap	0.582 (baseline 0.753)	computed above
N/P (practical, reversible)	measured cell cap 5.0282 Ah / cathode practical ~5.74 Ah (R1)	~0.88 (anode-limited by design)	r5_f1_1c_spme.json + R1 computation
Cell thickness	sum of layer thicknesses	200.8 um	cell/r5_f1_energy.json:thickness_m

Sheet: Process

Parameter	Formula	Value	Note
Positive areal density	thickness x (1-porosity) x density	164.0 g/m2	75.6e-6 x 0.665 x 3262
Negative areal density	85.2e-6 x 0.58 x 1657	81.9 g/m2	porosity 0.42 (F1)
Positive compaction density	density x (1-porosity) / 1000	2.169 g/cm3	divide by 1000
Negative compaction density	1657 x 0.58 / 1000	0.961 g/cm3	divide by 1000
Electrolyte fill amount	pore volume x 1.2 g/cm3 x fill factor 1.0	8.23 g	pore volume 6.855 cm3; literature density annotated
Formation recommendation	design recommended value	0.1C CC to 4.2 V, 25 C, 2 cycles	production value requires tuning